Chapter2 Electricity

I. Electrical energy and Heat energy

1. How much heat is absorbed by 3.50×10^2 g of water when it's heated from 15 °C to 85 °C?
2. How much heat in kJ is needed to raise the temperature of 60 g of water from 45 °C to 83 °C?
3. How much heat in joules must be added to 80 kg of aluminum at 30 °C to raise its temperature to130 °C?
4. **(a)** How much heat is required to raise the temperature of 300 mL of water from 20 °C to 35 °C?
   (b) How much heat is lost by the water as it cools back down to 20 °C?
5. How much heat does 45 g of aluminum give off as it cools from 100 °C to 30 °C? For aluminum C= 903 J/kg °C.
6. (a) Calculate the amount of energy needed to change the temperature of:
   (i) 500 g of water by 5 °C (specific heat capacity of water is 4200 J/kg °C)
   (ii) 100 g of aluminum from 100 °C to 300 °C (specific heat capacity of aluminum is 880 J/kg °C).
   (b) A 4 kg lump of iron is given 20 kJ of energy. It rises in temperature by 10 °C. Calculate the specific heat capacity of iron.
7. A heating coil has a resistance of 4 Ω and operates on 120V.
   a. What is the current in the coil while it is operating?
   b. What energy is supplied to the coil in 5 min?
   c. If the coil is immersed in an insulated container holding 20kg of water, what will be the increase in the temperature of the water? Assume that 100% of the heat is absorbed by the water.
8. An electric heater is rated at 1000W.
   a. How much energy is delivered to the heater in 2 hours?
   b. The heater is being used to heat a room containing 100 kg of air. If the specific heat of air is (1100 J/kg°C) and 70% of the thermal energy heats the air in the room, what is the change in air temperature?
9. The resistance of an electric stove element at operating temperature is 11 kΩ .
   a. 220V is applied across it. What is the current through the element?
   b. How much energy does the element use in 30 mins?
   c. The element is being used to heat a kettle containing 1.20 kg of water. Assume that 70% of the heat is absorbed by the water. What is its increase in temperature during the 30 s?
10. An electric heater is rated at 500W.
    a. How much energy is delivered to the heater in half an hour?
    b. The heater is being used to heat a room containing 50 kg of air. If the specific heat of air is 1.10 kJ/kg°C (1100 J/kg°C) and 50% of the thermal energy heats the air in the room, what is the change in air temperature?
11. A consumer requires an immersion heater for a tank containing 140 liters of Water. The water is to be heated from 6 °C to 66 °C. 3.5 hours. Calculate the nearest element size (in kilowatts) if the heating system is 90 percent efficient. Given as 1-liter weights 1kg and 1kWh = 3600000J.
12. A water heater has a volume of 10 litters and is fitted with a 2kW immersion heater. Calculate the time taken to raise the temperature of water from 4 °C to 46 °C. The water heater is 85 percent efficient.
13. A water heater takes 5hours to heat 180 litters of water from 5 °C to 60 °C. Calculate the efficiency of water if the element is 3kW.
14. A current was sent through a resistance wire of 10 ohms, which is fully immersed in 2 kg of water. At the end of 15 minutes, the rise in temperature was observed to be 60 °C. Determine the value of current.
15. An electric iron with a resistance of 60 Ω is crossed by a current of 4 A.
    a. What is voltage supplied from the battery?
    b. Calculate the electric power of an electric iron.
    c. Calculate the electrical energy dissipated from iron for 25 minutes.
16. An electric current of 2 A flows through a small piece of wire that is placed in water with a mass of 0.40 kg and a duration of 250 seconds, and the temperature of the water rises by 10 °C. Assume that the specific heat capacity of water 4200 J/kg°C. Calculate the resistance of the wire.
17. An electric iron with a resistor of 25 Ω and a power of 900 W is used for 2 hours every day.
    a. Calculate electric current across electric iron.
    b. Calculate the electric energy is dissipated by iron for 30 days.

c. What is the cost of using iron for a month if the electricity cost 1kWh is 720 riels?

18. The 2 kg of water is heated to the boiling point at a temperature of 20 °C. Water has a capacity of 4200 J/kg °C.
a. Calculate the heat energy needed to boil water.
b. In real-world application, an electric kettle with a power of 1600 W is used to heat water for 8 minutes and 45 seconds. Calculate the electric energy dissipated by the kettle under steady-state conditions.
c. Calculate the efficiency of a kettle.

19. An electric iron with a resistance of 24 Ω and a power of 600 W.
a. Determine the flow of electric current through iron.
b. Calculate the electrical energy in kilowatt-hours and joules.
c. Calculate the cost of electricity for 30 days if they used an electric iron for 2 hours per day and 1 kWh is 400 riels.

20. Calculate the heat energy dissipated in 10 seconds from a wire by a 2.09 Ω resistor when an electric current flows across 2.5 A.

II. Electric energy, heat liberated and light energy

21. A 6-V battery delivers 0.5A of current to an electric motor connected across its terminals.
a. What is the power rating of the motor?
b. How much energy does the motor use in 5.0 minutes? The current through the starter motor of a car connected to a 12-V battery is 210 A. What electric energy is delivered to the starter in 10 s?

22. A flashlight is connected across a 3 V difference in potential. The current through the lamp is 1.5 A.
a. What is the power rating of the lamp?
b. How much electric energy does the lamp convert in 10 min?

23. A lamp draws a 0.50 A from a 120-V generator.
a. How much power does the generator deliver?
b. How much energy does the lamp convert in 5 minutes?

24. A 15-Ω electric heater operates on a 120-V outlet.
a. What is the current through the heater?
b. How much energy is used by the heater in 30.0 seconds?
c. How much heat is liberated by the heater in this time?

25. A 30-Ω resistor is connected to a 60-V battery.
a. What is the current in the circuit?
b. How much energy is used by the resistor in 5 minutes?

26. A 100-W light bulb is 20% efficient. That means 20% of the electric energy is converted to light energy.
a. How many joules does the light bulb convert into light each minute it is in operation?
b. How many joules of heat does the light bulb produce each minute?

27. How much energy does a 60.0-W light bulb use in half an hour? If the light bulb is 12% efficient, how much heat does it generate during the half hour?

28. The new color television set draws 2.0A when operated on 120V.
a. How much power does the set use?
b. If the set is operated for an average of 7.0 hours per day, what energy in kWh does it consume per month (30days)?
c. At $0.08 per kWh, what is the cost of operating the set per month?

29. An electric space heater draws 15.0A from a 120-V source. It is operated, on the average, for 5.0h each day.
a. How much power does the heater use?
b. How much energy in kWh does it consume in 30 days?
c. At $0.08 per kWh, what does it cost to operate the heater in 30 days?

30. A digital clock has an operating resistance of 12 000 Ω and is plugged into a 115-V outlet.
a. How much current does it draw?
b. How much power does it use?
c. If the owner of the clock pays $0.09 per kWh, what does it cost to operate the clock for 30 days?